Software Engineering

Moscow Institute of Physics and Technology

01. Introduction and Environment

- 01.01 – 01.01 : Introduction. Overview. The C++ Programming Language. Programming Paradigms. Resources.
- 01.02 – 01.03 : Program Hello, World! Standard Template Library.

02. Language Core Basics

- 02.01 – 02.07 : Declarations and Definitions. Objects, Variables and Literals. Types. Type Aliases.
- 02.08 – 02.10 : Initialization. Type Inference. Type Conversions.
- 02.11 – 02.25 : Imperative and Structured Programming. Operators, Expressions and Statements.
- 02.26 – 02.35 : Memory. Addresses and Pointers. Arrays. Lvalue References.
- 02.36 – 02.45 : Functions. Procedural Programming. Algorithms. Hybrid Sort. Binary Search. Time Complexity.

03. Object Oriented Programming

- 03.01 – 03.06 : Object - Oriented Programming. Structures. Classes. Encapsulation.
- 03.07 – 03.14 : Class Relations. Patterns Attorney - Client and PassKey. Class Hierarchies. Inheritance.
- 03.15 – 03.22 : Dynamic Polymorphism. Abstraction. RunTime Type Identification.
- 03.23 – 03.28 : Rvalue References. Copy and Move Semantics. Pattern Copy - And - Swap. Rules of 0, 3, 4 and 5.
- 03.29 – 03.35 : Operator Overloading. Three - Way Comparison.

04. Generic Programming

- 04.01 – 04.08 : Generic Programming. Function and Class Templates. Template Specializations.
- 04.09 – 04.11 : Variadic Templates. Variadic and Fold Expressions.
- 04.12 – 04.14 : Forwarding References. Perfect Forwarding. Pattern SFINAE.
- 04.15 – 04.24 : Type Alias and Variable Templates. Metaprogramming. Constant Expressions. Tuples. Type Lists.
- 04.25 – 04.39 : Type Traits. Metafunctions. Concepts, Constraints and Requirements.

05. Software Engineering Patterns

- 05.01 – 05.06 : Patterns Builder, Abstract Factory, Prototype, Singleton and Noncopyable.
- 05.07 – 05.11 : Patterns Adapter, Bridge, Composite, Decorator and Facade.
- 05.12 – 05.16 : Patterns Memento, Observer, State, Strategy and Template Method. Policy - Based Programming.
- 05.17 – 05.24 : Static Polymorphism. Curiously Recurring Template Pattern. Pattern Mixin.

06. Projects and Libraries

- 06.01 – 06.04 : Preprocessing. Translation Units. Preprocessor Directives. Macros.
- 06.05 – 06.11 : Compilation, Assembly and Linking. Header, Source, Object and Executable Files. Pattern Pimpl.
- 06.12 – 06.20 : Namespaces. Modules. Libraries. Boost Library Collection.

07. Debugging and Profiling Tools

- 07.01 – 07.10 : Error Handling. Assertions. Terminations. Error Codes. Enumerations. Unions.
- 07.11 – 07.13 : Exceptions. Exception Safety Guarantees.
- 07.14 – 07.23 : Developer Tools. Backtracing, Debugging, Profiling, Logging, Testing and Microbenchmarking.

08. Applied Computations

- 08.01 – 08.09 : Bits and Bytes. Reflected Binary Gray Code. Bit Fields.
- 08.10 – 08.18 : Long Arithmetic. Karatsuba Fast Multiplication. Embedding Python.
- 08.19 – 08.25 : Floating - Point Numbers. Numerical Methods. Statistics. Complex Numbers. Fourier Transforms.
- 08.26 – 08.29 : Random Numbers. Monte - Carlo Methods.
- 08.30 – 08.39 : Clocks, Durations and Time Points. Timing. Calendars.

09. Memory Management

- 09.01 – 09.17 : Pattern RAII. Garbage Collectors. Smart Pointers. Iterators. Iterator Traits.
- 09.18 – 09.26 : Memory. Memory Segments. Data Alignment. Error Handling. Uninitialized Memory. Optionals.
- 09.27 – 09.36 : Allocators. Allocator Traits. Polymorphic Resources and Allocators.

10. Programming with Containers

- 10.01 – 10.21 : Sequential Containers. Container Adapters. Heaps. Circular Buffers. Patterns Proxy and Flyweight.
- 10.22 – 10.30 : Nested and Multidimensional Containers. Dynamic Programming.
- 10.31 – 10.33 : Linear Algebra. Matrices. Strassen Fast Multiplication. Decomposition Methods.
- 10.34 – 10.39 : Balanced Binary Search Trees. Sets. Associative Containers.
- 10.40 – 10.46 : Hash Functions. Hash Tables. Unordered Containers. Multiple Interfaces.

11. Programming with Algorithms

- 11.01 – 11.11 : Functors and Predicates. Lambda Expressions. Event - Driven Programming. Pattern Command.
- 11.12 – 11.15 : Pattern Visitor Implementations. Variants.
- 11.16 – 11.28 : Ranges. Algorithms. Iterator and Function Adapters. Views.
- 11.29 – 11.33 : Graphs. Breadth First Search. Depth First Search. Visitors. Dijkstra Shortest Paths.

12. Text Data Processing

- 12.01 – 12.12 : Locales. Facets. Encodings. Standard Unicode. Encoding Conversions. Strings.
- 12.13 – 12.26 : Regular Expressions. Grammars. Extended Backus - Naur Form. Parsers. Abstract Syntax Trees.

13. Input and Output Subsystems

- 13.01 – 13.15 : Streams. Flags. Manipulators. Stream Synchronization. Stream Buffers. Iterators.
- 13.16 – 13.23 : Filesystem. Serialization. Formats. Databases. Embedding PostgreSQL. Declarative Programming.

14. Parallel Programming

- 14.01 – 14.09 : Parallel Programming. Concurrency. Multitasking. Threads.
- 14.10 – 14.14 : Asynchronous Programming. Futures. Packaged Tasks. Promises. Algorithms. Execution Policies.
- 14.15 – 14.27 : Synchronization. Mutexes, Condition Variables, Semaphores, Latches and Barriers.
- 14.28 – 14.33 : Architectures. Processors, Chiplets and Cores.
- 14.34 – 14.35 : Execution Pipelines. Instructions. Memory Management Unit. Memory Hierarchy. Caches.
- 14.36 – 14.46 : Atomics. Cache Coherence Protocols. Memory Ordering. Spinlocks.
- 14.47 – 14.52 : Lock - Free Programming. Hazard Pointers. Pattern Disruptor. Thread Pools.
- 14.53 – 14.60 : Processes. Standard POSIX. Inter - Process Communications. Shared Memory. Synchronization.

15. Distributed Network Systems

Work in progress ...